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Editor-in-Chief
IEEE Access

Dear Editor-in-Chief,

I am pleased to submit the manuscript entitled

*Predictive Energy Management of an AI/HPC Data Center with BESS/UPS:
A Comparative Evaluation of Heuristic, Myopic Optimization, and MPC*

for consideration as a Research Article in *IEEE Access*.

This work presents an applied, comparative energy management framework for AI/HPC data centers with BESS providing UPS support. Rather than introducing a new MPC theory result, the paper provides a reproducible simulation protocol that separates forecast inputs from the optimizer and compares, under the same plant and metrics: (i) a reactive heuristic, (ii) a foresight-enabled heuristic, (iii) a one-step myopic QP, and (iv) an event-anticipating MPC-QP controller.

The study focuses on AI-typical demand steps and source-limit saturation, and includes foresight ablations, multi-burst profiles, forecast-error tests, and sensitivity analyses. The results show when predictive BESS/UPS coordination removes converter-limited unserved-power transients, and also delimit the physical role of storage—including cases in which mistimed foresight can worsen the peak relative to reactive control.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors approve this submission. No conflict of interest is declared. This work is part of an M.Sc. research project at UFMS; no external grant number is reported.

Thank you for considering our manuscript. I look forward to your editorial assessment.

Sincerely,

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